MATH 201 Your name: ____________________

Learning target AD5 and AD4, version 1

AD5: Draw annotated number lines and apply Derivative Tests to classify the critical points as local extrema.
AD4: Use the Extreme Value Theorem to find the global maximum and minimum values of a continuous function on a closed interval.

Consider the function $f(x) = 2x^3 + 24x^2 - 54x + 7$.

1. (AD5) Find the critical points using algebra, and draw a sign chart (aka labeled number line).

2. (AD4) Restricting the domain of this function to the interval $[-8, 2]$, what are the absolute maximum and absolute minimum values of this function, and where do they occur?

3. (AD4) What if we change the domain to $[-10, 2]$?